# BARATHKUMAR V

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## Summary

Motivated B.Tech graduate in Artificial Intelligence and Data Science with strong foundations in Data Structures and Algorithms, object-oriented programming (OOP), and the software development lifecycle (SDLC). Proficient in Java and Python with hands-on experience building full-stack and machine learning applications. Seeking a software engineering role at TCS to contribute to large-scale software delivery, applying technical depth, agile teamwork, and a commitment to continuous improvement.

## Education

**P.A. College of Engineering and Technology** 2022 – 2026
*B.Tech in Artificial Intelligence and Data Science CGPA: 7.72* *Pollachi, Tamil Nadu*

**Blue Bird Matric Higher Secondary School** 2022
*HSC – Percentage: 76.1%* *Palladam, Tamil Nadu*

**Blue Bird Matric Higher Secondary School** 2020
*SSLC – Percentage: 88.4%* *Palladam, Tamil Nadu*

## Technical Skills

**Programming Languages:** Java, Python, C, JavaScript, SQL

**Core CS Concepts:** Data Structures and Algorithms, Object-Oriented Programming (OOP), Database Management Systems (DBMS)

**Web Development:** HTML, CSS, JavaScript, REST API, Flask, Streamlit

**Machine Learning / AI Libraries:** Scikit-learn, NumPy, Pandas, Matplotlib, OpenCV, NLTK, TF-IDF, SVM

**Developer Tools:** Git, GitHub, VS Code, Jupyter Notebook

## Work Experience

**Jyesta Corporate Entity** Aug 2025 – Oct 2025
*Artificial Intelligence Intern*

- Designed and implemented AI-driven modules, integrating them into existing software pipelines.
- Wrote clean, maintainable Python code for AI workflows and automated data processing tasks.
- Collaborated with cross-functional teams using agile practices to deliver software components on schedule.

**CLOB Groups** Nov 2024 – Dec 2024
*Web Development Intern* *Coimbatore, Tamil Nadu*

- Engineered responsive, cross-browser-compatible web pages using HTML, CSS, and JavaScript.
- Implemented interactive UI components and optimized page performance for faster load times.
- Participated in the full software development lifecycle from requirement gathering to deployment.

**Live Stream Technology** Jun 2024 – Jul 2024
*Python and Data Science Intern* *Coimbatore, Tamil Nadu*

- Developed data processing scripts and automation pipelines using NumPy, Pandas, and Matplotlib.
- Built reusable Python modules to streamline repetitive data transformation tasks.
- Applied software best practices including modular design and version control using Git and GitHub.

## Projects

**Hands-Free Virtual Mouse using Computer Vision** 2025
*Python, OpenCV, PyAutoGUI*

- Engineered a gesture-recognition system using OpenCV with contour detection and convex hull analysis.
- Designed a low-latency event pipeline that maps real-time hand gestures to OS-level mouse controls.
- Applied object-oriented programming (OOP) principles to reduce hardware dependency for system interaction, improving accessibility.

**AI Chatbot for User Support** 2025
*Python, Natural Language Processing (NLP), Flask*

- Architected and developed an end-to-end intelligent chatbot application using Python and NLP techniques.
- Implemented TF-IDF vectorization and intent classification algorithms to achieve high-accuracy responses.
- Iterated on model performance through custom dataset training and automated feedback loops.

**Product Review Sentiment Classifier** 2025
*Python, Support Vector Machine (SVM), Streamlit*

- Built and deployed a full-stack sentiment analysis application with an SVM-based text classification engine.
- Engineered an NLP preprocessing pipeline including tokenization, stemming, and TF-IDF vectorization.
- Delivered a real-time web interface using Streamlit for instant predictions on user input.

**Machine Learning Real Estate Price Estimator** 2024
*Python, Scikit-learn, Streamlit*

- Developed a regression-based prediction system with full exploratory data analysis (EDA), feature engineering, and preprocessing.
- Automated hyperparameter tuning using GridSearchCV across multiple machine learning algorithms.
- Deployed a production-ready interactive web application using Streamlit for end-user predictions.

**Weather Dashboard** 2024
*HTML, CSS, JavaScript, REST API*

- Developed a dynamic, responsive single-page application (SPA) using HTML, CSS, and vanilla JavaScript.
- Integrated the OpenWeatherMap REST API with asynchronous fetch calls for real-time data rendering.
- Implemented dynamic theming and conditional UI rendering based on live weather state.

## Personal Details

**Date of Birth:** 22 September 2004
**Gender:** Male
**Nationality:** Indian
**Location:** Palladam, Tamil Nadu, India
**Languages Known:** English, Tamil
**Hobbies:** Reading, Music, Travel